

- I selected a subset of clinical observation data from the Synthea EHR dataset focusing on BMI, HbA1c, glucose, cholesterol, and diabetes-related indicators to analyze patient population health trends over time.
- The raw FHIR JSON observation data was flattened and transformed into analysis-ready tabular datasets using Python Pandas. Nested JSON structures such as observation coding and blood pressure components were extracted and normalized for dashboard analysis.
- The dashboard narrative focuses on longitudinal metabolic health analysis, highlighting obesity trends, diabetes risk stratification, observation volume, and multi-metric clinical progression across years.
- One of the most significant findings was the high proportion of overweight and prediabetic patient observations, indicating elevated metabolic health risk patterns within the selected cohort.
- If developed further for a production healthcare environment, the dashboard could be enhanced with patient demographics, encounter-level drilldowns, geographic segmentation, and real-time clinical alert monitoring.